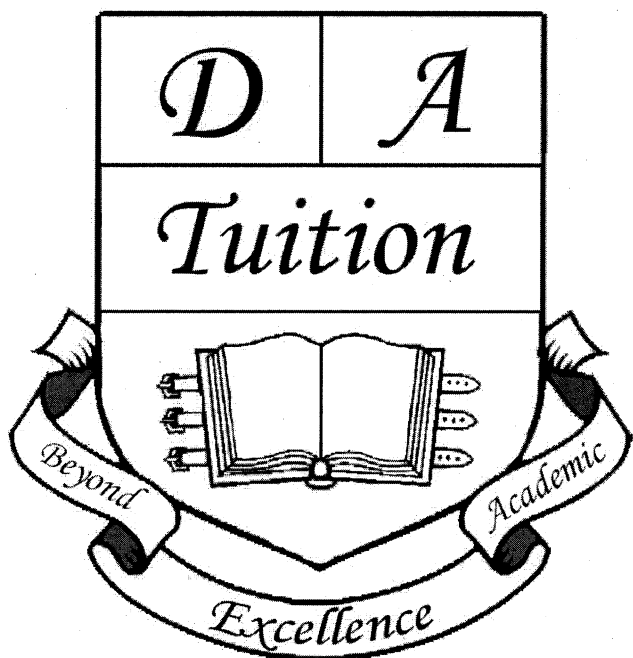




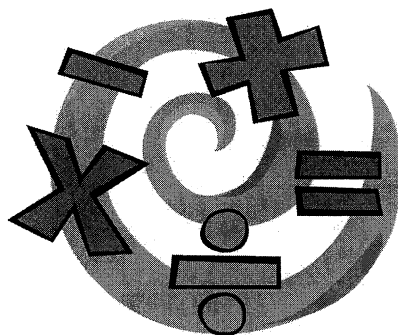
2unit

Mathematics

Term ____ Week ____

CALCULUS 1

introductory calculus

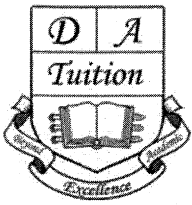


Name: Teacher's Copy

Mark:

Percentage:

Grade:



Calculus 1



Part I: The Derivative of x^n

1. Differentiate (with respect to x)

(a) 6

$$0$$

(b) $20 - 4x$

$$-4$$

(c) $\frac{x}{5}$

$$\frac{1}{5}$$

(d) $\frac{8x+9}{2} = 4x + \frac{9}{2}$

$$4$$

(e) $4x^2 - 9x + 6$

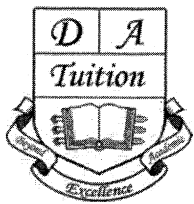
$$8x - 9$$

(f) $5 + 6x - 3x^2$

$$6 - 6x$$

(g) $\frac{7x^2 + 3x}{x} = 7x + 3$

$$7$$

**Calculus 1**

(h) $\frac{8x^5 - 10x^6}{2x^2} = 4x^3 - 5x^4$

$$12x^2 - 20x^3$$

(i) $x(x-9) = x^2 - 9x$

$$2x - 9$$

(j) $\frac{1}{2}x^4 - 5x^3 - 2x + 4$

$$4x^3 - 15x^2 - 2$$

(k) $\frac{3}{4}x^4 + \frac{x^3}{3} - \frac{1}{2}x^2$

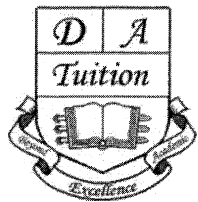
$$3x^3 + x^2 - x$$

(l) πx^2

$$2\pi x$$

(m) $\frac{4}{3}\pi x^3$

$$4\pi x^2$$

**Calculus 1**

(n) $cx^2 + dx + e$

$$2cx + d$$

(o) $3x(x^3 - 9x^2 + 4x - 2) \quad 3x^4 - 27x^3 + 12x^2 - 6x$

$$12x^3 - 81x^2 + 24x - 6$$

(p) $\frac{12x^5 - x^4 + 24x^3}{6x^2}$

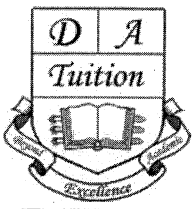
$$6x^3 - \frac{x}{3} + 4$$

(q) $(2x + 5)(x + 3)$

$$4x + 11$$

2. (a) Evaluate $f'(2)$ if
 $f(x) = 2x^3 - 5x^2 + 1$

$$4$$



Calculus 1



- (b) Evaluate $f'(2)$ if

$$f(x) = x^4 + \frac{1}{2}x^3$$

38

- (c) Evaluate $f(5) + 2f'(-1)$ if

$$f(x) = 3x^2 + 4x + 2$$

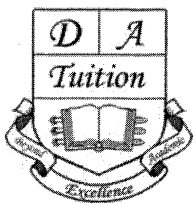
93

- (d) Given $f(x) = ax^2 + 5x - 9$
evaluate a if $f'(3) = 47$.

7

- (e) Given $f(x) = 4x^2 + 3x + 2$,
evaluate x if $f'(x) = -13$.

-2



Calculus 1



- (f) At what point on the curve
 $f(x) = x^3 - 2x^2 - 4x + 2$ is
 $f'(x) = 0$? (for $x > 0$)

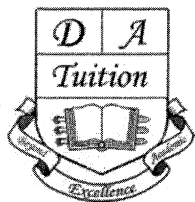
$(2, -6)$

- (g) Given $f(x) = 6x + 24$ find x such
that $f(x) = f'(x)$.

-3

- (h) Given $f(x) = 5x^2 - 8x - 16$ find
 x such that $f(x) = f'(x)$.

$-\frac{2}{5}$ or 4



Calculus 1



Part 2: The Derivative of x^{-n}

1. Differentiate (write answers with positive indices).

(a) x^{-2}

$$-\frac{2}{x^3}$$

(b) x^{-5}

$$-\frac{5}{x^6}$$

(c) $3x^{-4}$

$$-\frac{12}{x^5}$$

(d) $\frac{1}{2}x^{-6}$

$$-\frac{3}{x^7}$$

(e) $\frac{2}{3}x^{-3}$

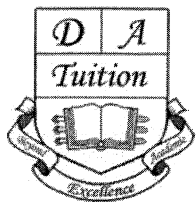
$$-\frac{2}{x^4}$$

(f) $\frac{x^{-1}}{4}$

$$-\frac{1}{4x^2}$$

(g) $\frac{3}{4}x^{-2}$

$$-\frac{3}{2x^3}$$



Calculus 1



(h) $\frac{7}{x}$

$$-\frac{7}{x^2}$$

(i) $\frac{-8}{x^2}$

$$\frac{16}{x^3}$$

(j) $\frac{-2}{5x^5}$

$$\frac{2}{x^6}$$

(k) $\frac{1}{2x^4}$

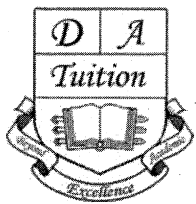
$$-\frac{2}{x^5}$$

(l) $3x^2 + \frac{4}{x}$

$$6x - \frac{4}{x^2}$$

(m) $\frac{5x^4 - 3x^2}{x^3}$

$$5 + \frac{3}{x^2}$$



Calculus 1



(n) $\frac{8x^6 + 2x^3}{4x^4}$

$$4x - \frac{1}{2x^2}$$

(o) $16x + \frac{9}{x}$

$$16 - \frac{9}{x^2}$$

(p) $2\pi x^2 + \frac{36}{x}$

$$4\pi x - \frac{36}{x^2}$$

Part 3: The Derivative of $x^{m/n}$

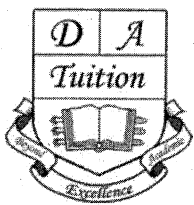
1. Differentiate (write answers in surd form):

(a) $x^{\frac{1}{2}}$

$$\frac{1}{2\sqrt{x}}$$

(b) $4x^{\frac{1}{2}}$

$$\frac{2}{\sqrt{x}}$$



Calculus 1



(c) $\frac{1}{3}x^{\frac{1}{2}}$

$$\frac{1}{6\sqrt{x}}$$

(d) $x^{\frac{1}{3}}$

$$\frac{1}{3\sqrt[3]{x^2}}$$

(e) $9x^{\frac{1}{3}}$

$$\frac{3}{3\sqrt{x^2}}$$

(f) $2x^{\frac{1}{3}}$

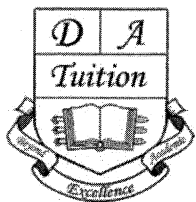
$$\frac{2}{3\sqrt[3]{x^2}}$$

(g) $8x^{\frac{3}{4}}$

$$\frac{6}{4\sqrt{x}}$$

(h) $12x^{\frac{1}{4}}$

$$\frac{3}{4\sqrt{x^3}}$$



Calculus 1



(i) $x^{\frac{1}{5}}$

$$\frac{1}{5\sqrt[5]{x^4}}$$

(j) $10x^{\frac{2}{5}}$

$$\frac{4}{5\sqrt[5]{x^3}}$$

(k) $8\sqrt{x}$

$$\frac{4}{\sqrt{x}}$$

(l) $6\sqrt[3]{x}$

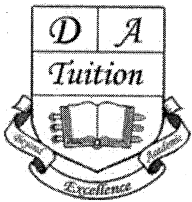
$$\frac{2}{3\sqrt[3]{x^2}}$$

(m) $9\sqrt[3]{x^2}$

$$\frac{6}{3\sqrt[3]{x}}$$

(n) $4\sqrt[4]{x^3}$

$$\frac{3}{4\sqrt[4]{x}}$$

**Calculus 1**

(o) $x^{-\frac{1}{2}}$

$$-\frac{1}{2\sqrt{x^3}}$$

(p)* $-6x^{-\frac{2}{3}}$

$$\begin{aligned} &= -6x^{-\frac{2}{3}} x^{-\frac{5}{3}} \\ &= 4x^{-\frac{5}{3}} \\ &= \frac{4}{3\sqrt{x^5}} \end{aligned}$$

(q) $5x^2 + \frac{2}{x} + x^{\frac{1}{2}}$

$$10x - \frac{2}{x^2} + \frac{2}{\sqrt{x}}$$

(r) $6x^3 + \frac{5}{x^2} + 3x^{\frac{1}{3}}$

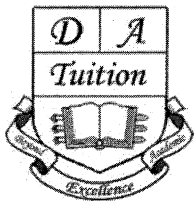
$$18x^2 - \frac{10}{x^3} + \frac{1}{3\sqrt{x^2}}$$

(s) $4x^4 + 6\sqrt{x}$

$$16x^3 + \frac{3}{\sqrt{x}}$$

(t) $\frac{1}{2}x^6 - \frac{2}{x^3} + \frac{1}{\sqrt{x}}$

$$3x^5 + \frac{6}{x^4} - \frac{1}{2\sqrt{x^3}}$$



(u) $\frac{9}{\sqrt[3]{x}}$

$$-\frac{3}{\sqrt[3]{x^4}}$$

(v)* $y = x^{4.5} \sqrt{x} = x^{\frac{9}{2}} x^{\frac{1}{2}}$

$$= x^5$$

$$y' = 5x^4$$

(w) $7x^{\frac{2}{3}} \cdot \sqrt[3]{x}$

$$7$$

(x) $\frac{8x}{\sqrt{x}}$

$$\frac{4}{\sqrt{x}}$$

Part 4: Function of a Function Rule

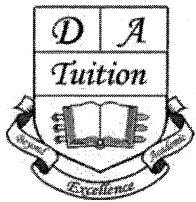
1. Differentiate:

(a) $(x+2)^5$

$$5(x+2)^4$$

(b) $(3x+6)^7$

$$21(3x+6)^6$$

**Calculus 1**

(c) $(8-4x)^6$

$$-24(8-4x)^5$$

(d) $(5x^2-3)^4$

$$40x(5x^2-3)$$

(e) $\left(\frac{x}{2}+3\right)^8$

$$4\left(\frac{x}{2}+3\right)^7$$

(f) $(x+5)^{-1}$

$$-\frac{1}{(x+5)^2}$$

(g) $2(x-4)^{-1}$

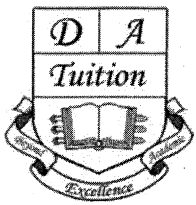
$$-\frac{2}{(x-4)^2}$$

(h) $(3x+2)^{-1}$

$$-\frac{3}{(3x+2)^2}$$

(i) $(x+4)^{-2}$

$$-\frac{2}{(x+4)^3}$$



Calculus 1



(j) $5(x-7)^{-2}$

$$\frac{-10}{(x-7)^3}$$

(k) $(9-3x)^{-2}$

$$\frac{6}{(9-3x)^3}$$

(l) $5(2x+4)^{-3}$

$$\frac{-30}{(2x+4)^4}$$

(m) $-6(4x-7)^{-4}$

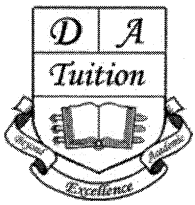
$$\frac{96}{(4x-7)^5}$$

(n) $\frac{1}{(5x+2)}$

$$\frac{-5}{(5x+2)^2}$$

(o) $\frac{3}{6x-5}$

$$\frac{-18}{(6x-5)^2}$$



Calculus 1



(p) $\frac{1}{(2x+3)^2}$

$$\frac{-4}{(2x+3)^3}$$

(q) $\frac{4}{(5x+3)^3}$

$$\frac{-60}{(5x+3)^4}$$

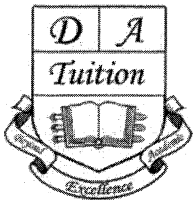
(r) $\frac{2}{(2x+1)^4}$

$$\frac{-16}{(2x+1)^5}$$

2. Differentiate (answers in surd form):

(a) $(2x+8)^{\frac{1}{2}}$

$$\frac{1}{\sqrt{2x+8}}$$



Calculus 1



(b) $\sqrt{x+6}$

$$\frac{1}{2\sqrt{x+6}}$$

(c) $\sqrt{3x+4}$

$$\frac{3}{2\sqrt{3x+4}}$$

(d) $\sqrt{9-8x}$

$$-\frac{4}{\sqrt{9-8x}}$$

(e) $\sqrt[3]{x-5}$

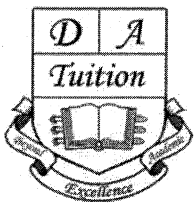
$$\frac{1}{3\sqrt[3]{(x-5)^2}}$$

(f) $4\sqrt[4]{(3x+1)}$

$$\frac{3}{4\sqrt[4]{(3x+1)^3}}$$

(g) $\sqrt{25-x^2}$

$$-\frac{x}{\sqrt{25-x^2}}$$



Calculus 1



(h) $(10x+3)^{-\frac{1}{2}}$

$$-\frac{5}{\sqrt{(10x+3)^3}}$$

(i) $(20-6x)^{-\frac{1}{3}}$

$$\frac{2}{3\sqrt{(20-6x)^4}}$$

(j) $\frac{1}{\sqrt{2x+5}}$

$$-\frac{1}{\sqrt{(2x+5)^3}}$$

(k)* $\frac{9}{\sqrt[3]{12x-4}} = 9(12x-4)^{-\frac{1}{3}}$

$$= -3(12x-4)^{-\frac{4}{3}} \times 12$$

$$= -36(12x-4)^{-\frac{4}{3}} = \frac{-36}{\sqrt[3]{(12x-4)^4}}$$

(l) $\frac{2}{5\sqrt{10x+1}}$

$$-\frac{2}{5\sqrt{(10x+1)^3}}$$

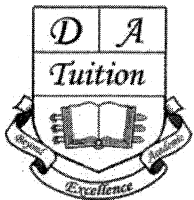
Part 5: Product Rule

1. Complete the factorisation:

(a) $(2x+1)^4 = (2x+1)^3 (2x+1)$

(b) $(x+5)^{\frac{1}{2}} = (x+5)^{-\frac{1}{2}} (x+5)$

(c) $(x-6)^{\frac{1}{3}} = (x-6)^{-\frac{2}{3}} (x-6)$



2. Factorise fully:

(a) $24x(x+2)^3 + 6(x+2)^4$

$$6(5x+2)(x+2)^3$$

(b)

$6(3x+5)^2(2x+3)^2 + 6(2x+3)^3(3x+5)$

$$6(3x+5)(5x+8)(2x+3)^2$$

(c)* $5x(x-3)^{-\frac{1}{2}} + 10(x-3)^{\frac{1}{2}} = 5(x-3)^{-\frac{1}{2}} [x(x-3)^0 + 2(x-3)]$

$$\begin{aligned} &= 5(x-3)^{-\frac{1}{2}} (x + 2x - 6) \\ &= 5(3x-6)(x-3)^{-\frac{1}{2}} \\ &= 15(x-2)(x-3)^{-\frac{1}{2}} \end{aligned}$$

3. Use the product rule to differentiate each of the following:

(a) $(x-2)(x^2+3)$

$$3x^2 - 4x + 3$$

(b) $(x+5)(x^2+3x)$

$$3x^2 + 16x + 15$$



(c) $(x^3 + 4x)(6 + 2x)$

$$8x^3 + 18x^2 + 16x + 24$$

(d)* $x\sqrt{1-x^2}$ $u = x$ $v = (1-x^2)^{\frac{1}{2}}$

$$u' = 1 \quad v' = \frac{1}{2} (1-x^2)^{-\frac{1}{2}} \cdot -2x$$

$$= \frac{-x}{\sqrt{1-x^2}}$$

$$y' = vu' + uv'$$

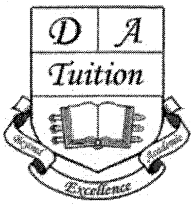
$$= (1-x^2)^{\frac{1}{2}} + \frac{-x^2}{\sqrt{1-x^2}}$$

(e) $5x(x+1)^3$

$$5(x+1)^2(4x+1)$$

(f) $(2x+4)^3(6x+1)$

$$6(8x+5)(2x+4)^2$$



Calculus 1



(g) $3x\sqrt{x+2}$

$$\frac{9x+12}{2\sqrt{x+2}}$$

(h) $(x^4+2)(3x+1)$

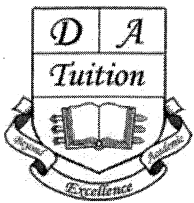
$$15x^4 + 4x^3 + 6$$

(i) $(2x+3)^4(6x)$

$$6(10x+3)(2x+3)^3$$

(j) $(5+x)\sqrt{25-x^2}$

$$\frac{-2x^2 - 5x + 25}{\sqrt{25-x^2}}$$



(k) $3x \sqrt[3]{x+4}$

$$\frac{4x+12}{\sqrt[3]{(x+4)^2}}$$

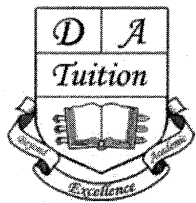
- 4.* Given $f(x) = (2x+8)(x+2)^3$, evaluate x such that $f'(x) = 0$. $u = 2x+8$ $v = (x+2)^3$

$$u' = 2 \quad v' = 3(x+2)^2$$

$$\begin{aligned} y' &= (2x+8)(3)(x+2)^2 + 2(x+2)^3 \\ &= 2(x+2)^2 [(x+4) \times 3 + 2(x+2)] \\ &= 2(x+2)^2 (3x+12 + 2x+4) \\ 0 &= 2(x+2)^2 (5x+16) \\ x &= -2, -\frac{16}{5} \end{aligned}$$

5. Given $f(x) = (2x-4)^2(3x+6)^2$, solve $f'(x) = 0$.

$$x = -2, 0, 2$$



Part 6: Quotient Rule

1. Differentiate:

(a) $\frac{9x}{8x+2}$

$$\frac{18}{(8x+2)^2}$$

(b) $\frac{2x+5}{x+3}$

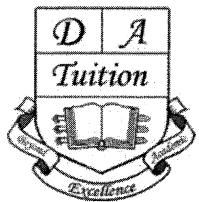
$$\frac{1}{(x+3)^2}$$

(c) $\frac{3x-4}{2x+7}$

$$\frac{29}{(2x+7)^2}$$

(d) $\frac{x}{x^2+3}$

$$\frac{3-x^2}{(x^2+3)^2}$$



Calculus 1

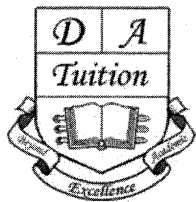


(e) $\frac{6x^2}{2x+5}$

$$\frac{12x^2 + 60x}{(2x+5)^2}$$

(f) $\frac{3x}{(x+1)^2}$

$$\frac{3(1-x)}{(1+x)^3}$$



Calculus 1

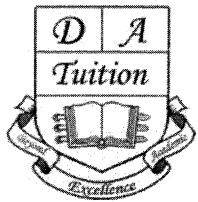


2. Find the equation of the tangent to the
curve $y = \frac{x+5}{x-4}$ at the point where $x = 7$.

$$\therefore x + y - 11 = 0$$

3. Find the equation of the normal to the
curve $y = \frac{8x}{x-2}$ at the point where $x = 4$.

$$x - 4y + 60 = 0$$



4. Find the points on the curve $y = \frac{9x}{1+x^2}$
where $f'(x) = 0$.

$(-1, -4.5)$ and $(1, 4.5)$

Part 7: Limits

Evaluate:

1. $\lim_{x \rightarrow 3} (12 - 2x)$

$$12 - 2(3) = 6$$

2. $\lim_{x \rightarrow 5} (x^2 + 3x - 2)$

$$= 5^2 + 3(5) - 2 = 38$$

3. $\lim_{x \rightarrow 0} \frac{x^2 + 7x}{x}$

$$\begin{aligned} &= \lim_{x \rightarrow 0} (x + 7) \\ &= 7 \end{aligned}$$

4. $\lim_{x \rightarrow -10} \frac{x^2 - 100}{x + 10}$

$$\begin{aligned} &= \lim_{x \rightarrow -10} \frac{(x-10)(x+10)}{(x+10)} \\ &= \lim_{x \rightarrow -10} (x-10) = -20 \end{aligned}$$

Calculus 1



5. (a) $\lim_{x \rightarrow -1} \frac{x^2 + 15x + 14}{x + 1} = \lim_{x \rightarrow -1} \frac{(x+14)(x+1)}{(x+1)}$

$$= \lim_{x \rightarrow -1} x + 14$$

$$= 13$$

(b) The graph of $y = \frac{x^2 + 15x + 14}{x + 1}$

is the line $y = x + 14$

with a discontinuity at the point

$(-1, 13)$

6. $\lim_{x \rightarrow 3} \frac{x^3 - 27}{x - 3}$

$$= \lim_{x \rightarrow 3} \frac{(x-3)(x^2 + 3x + 9)}{(x-3)}$$

$$= \lim_{x \rightarrow 3} x^2 + 3x + 9$$

$$= 3^2 + 3(3) + 9$$

$$= 27$$

7. $\lim_{x \rightarrow \infty} \frac{1}{x^2 + 4}$

$$\lim_{x \rightarrow \infty} \frac{\frac{1}{x^2}}{\frac{x^2}{x^2} + \frac{4}{x^2}}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{1}{x^2}}{1 + \frac{4}{x^2}}$$

$$= \frac{0}{1+0} = 0$$

8. $\lim_{x \rightarrow \infty} (2 + \frac{1}{x})$

$$= 2 + 0$$

$$= 2$$

9. $\lim_{x \rightarrow \infty} \frac{x-2}{x-3}$

$$\lim_{x \rightarrow \infty} \frac{\frac{x}{x} - \frac{2}{x}}{\frac{x}{x} - \frac{3}{x}}$$

$$= \lim_{x \rightarrow \infty} \frac{1 - \frac{2}{x}}{1 - \frac{3}{x}}$$

$$= \frac{1-0}{1-0} = 1$$



Part 8: First Principles

1. Find the gradient of the secant joining the points A(4, 35) and B(6, 75) on the parabola $y = 2x^2 + 3$.

$$m = \frac{75 - 35}{6 - 4} = 20$$

2. Find the gradient of the chord joining the points whose x values are -2 and 5 on the parabola $y = x^2 + 6x + 2$.

$$\text{at } x = -2, y = (-2)^2 + 6(-2) + 2 = -6$$

$$x = 5, y = 5^2 + 6(5) + 2 = 57$$

$$m = \frac{57 - (-6)}{5 - (-2)} = \frac{63}{7} = 9$$

3. (a) Simplify $\frac{h^2 + 2xh + 2h}{h} = \frac{h^2}{h} + \frac{2xh}{h} + \frac{2h}{h}$

$$h + 2x + 2$$

- (b) Evaluate $\lim_{h \rightarrow 0} \frac{h^2 + 2xh + 9h}{h}$

$$= \lim_{h \rightarrow 0} h + 2x + 9$$

$$= 2x + 9$$



4. Given $f(x) = x^2 + 8x + 7$:

(a) $f(x+h) = (x+h)^2 + 8(x+h) + 7$

$$= x^2 + 2xh + h^2 + 8x + 8h + 7$$

(b) $f(x+h) - f(x) = x^2 + 2xh + h^2 + 8x + 8h + 7 - (x^2 + 8x + 7)$

$$= x^2 + 2xh + h^2 + 8x + 8h + 7 - x^2 - 8x - 7$$

$$= 2xh + h^2 + 8h$$

(c) Evaluate: $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$$= \lim_{h \rightarrow 0} \frac{2xh + h^2 + 8h}{h}$$

$$= \lim_{h \rightarrow 0} (2x + h + 8) = 2x + 8$$

5. Using first principles find an expression for the gradient of the tangent to the curve $f(x) = 2x^2 - 5x + 3$.

$$= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2(x+h)^2 - 5(x+h) + 3 - (2x^2 - 5x + 3)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2x^2 + 4xh + 2h^2 - 5x - 5h + 3 - 2x^2 + 5x - 3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{4xh + 2h^2 - 5h}{h}$$

$$= \lim_{h \rightarrow 0} (4x + 2h - 5)$$

$$= 4x - 5$$



6. Differentiate $f(x) = 6x - x^2$ from first principles.

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{6(x+h) - (x+h)^2 - (6x - x^2)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\cancel{6x} + 6h - \cancel{x^2} - 2xh - h^2 - \cancel{6x} + \cancel{x^2}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{6h - 2xh - h^2}{h} \\
 &= \lim_{h \rightarrow 0} 6 - 2x - h \\
 &= 6 - 2x
 \end{aligned}$$

7. Find, from first principles, the gradient of the curve $f(x) = x^2 + 3x$ at the point $(8, 88)$.

$$\begin{aligned}
 y' &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{(x+h)^2 + 3(x+h) - (x^2 + 3x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\cancel{x^2} + 2xh + h^2 + \cancel{3x} + 3h - \cancel{x^2} - \cancel{3x}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{2xh + h^2 + 3h}{h} \\
 &= \lim_{h \rightarrow 0} 2x + h + 3 \\
 y' &= 2x + 3 \\
 \text{at } x=8, y' &= 2(8) + 3 \\
 &= \underline{\underline{19}}
 \end{aligned}$$